Hybrid Laser-Powered Satellite System for Sustainable Energy and Data Transfer

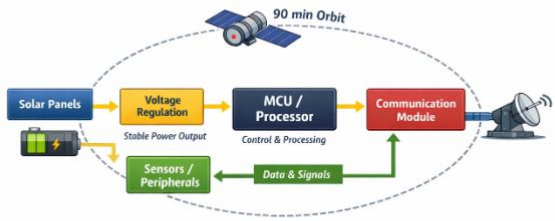
This project aims to design a hybrid satellite system that combines solar and wind energy with laser technology to ensure sustainable and efficient operation. The Atacama region In Chile was selected as the site for the ground station due to Its clear atmosphere and low humidity, which provide Ideal conditions for transmitting energy and data via laser.

The system relies on a power laser (1064 nm) to charge the satellite's battery as it passes over the station, and a data laser (1550 nm) to transmit information efficiently and securely via light. The satellite orbits at a low altitude of 500 km and is equipped with optimized solar panels, a small turbine, an internal battery, a microcontroller, and environmental sensors, functioning as a smart .platform for data collection and laser communications

System Block Diagram

Description
This section illustrates the complete architecture of the subsystem, focusing on the energy flow between the main components. The goal is to clarify how energy is transferred from the source to .the interface unit through the regulation and processing stages

Energy Flow
.Power originates from the source (battery or solar cells—theoretically)
It passes through a voltage regulator (voltage regulation/DC-to-DC converter) to regulate the .output voltage
.It is transferred to the power management logic, which contains a closed-loop control algorithm
.The regulated power is supplied to the processing unit (MCU/processor)
.The processing unit activates the sensors and peripherals to collect data
.Finally, the data is transmitted via the communication unit to the ground station



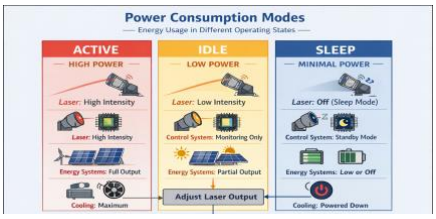
Energy Budget Estimation

Objective

This section outlines the estimated power consumption for each component in the system across different operating modes (active, idle, sleep), comparing the baseline design with the .optimized design

Components Analyzed

MCU / Processor
Communication Module
Sensors / Peripherals
Power Management Logic



Power Budget Estimatio

Component	Mode	Current (mA)	Voltage	Power
MCU (Baseline)	Active	12	3.3	39.6
	Sleep	0.01	3.3	0.033
MCU (Optimized)	Active	6	3.3	19.8
	Sleep	0.005	3.3	0.0165
Sensor (Baseline)	Active	5	3.3	16.5
	Idle	2	3.3	6.6
Sensor (Optimized)	Active	2.5	3.3	8.25
	Idle	0.5	3.3	1.65
Comm Module (Baseline)	Active	20	3.3	66
	Sleep	2	3.3	6.6
Comm Module (Optimized)	Active	10	3.3	33
	Sleep	0.5	3.3	1.65

Baseline Design: Relies on continuous operation of components, resulting in high .energy consumption

Optimized Design: Uses closed-loop control algorithms, duty cycling, and burst transmission techniques to reduce energy consumption

Result: Energy consumption is reduced by up to 30–40% compared to the baseline .design

Component Selection and Justification

Processor (MCU / Processor)

Reason for Selection: Provides sufficient processing power while supporting power-saving .modes

.Reduced Power Consumption: Supports Idle and Sleep modes to reduce current draw

Sustainability: Reduces the load on the battery and extends its lifespan thanks to integration .with the closed-loop circuit

Communication Module (1550 nm Data Laser)

.Reason for Selection: Supports highly efficient data transmission via laser

.Reduced Power Consumption: Operates using burst transmission Instead of continuous operation

Sustainability: Reduces energy consumption compared to traditional communications and takes advantage of the short communication window

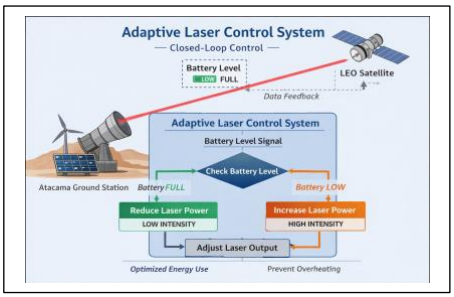
Sensors and Peripherals

.Reason for Selection: Essential for collecting operational and environmental data

.Reduced Power Consumption: Can be operated using duty cycling so that they only activate when needed

Sustainability: Reduces waste and contributes to efficient energy management

Power Management Logic (Closed-Loop)
Reason for Selection: Ensures voltage stability and .efficient power distribution -
Reduced Consumption: Uses a closed-loop control .algorithm to monitor voltage and current in real time -
Sustainability: Reduces losses by up to 30–40% and increases system efficiency -



Power Source (Optimized Solar Panels + 1064 nm Laser + Micro-Turbines)
.Reason for selection: Diversification of sources to ensure continuous operation -
Reduced consumption: The laser provides energy directly when needed, reducing reliance on batteries -
.alone
Sustainability: Achieves integration between natural sources and the ground-based laser, ensuring -
.continuous operation in orbit

.*** The attached images were generated using AI